



Uncertainty-aware image classification on 3D CT lung

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ABSTRACT

Early detection is crucial for lung cancer to prolong the patient's survival. Existing model architectures used in such systems have shown promising results. However, they lack reliability and robustness in their predictions and the models are typically evaluated on a single dataset, making them overconfident when a new class is present. With the existence of uncertainty, uncertain images can be referred to medical experts for a second opinion. Thus, we propose an uncertainty-aware framework that includes three phases: data preprocessing and model selection and evaluation, uncertainty quantification (UQ), and uncertainty measurement and data referral for the classification of benign and malignant nodules using 3D CT images. To quantify the uncertainty, we employed three approaches; Monte Carlo Dropout (MCD), Deep Ensemble (DE), and Ensemble Monte Carlo Dropout (EMCD). We evaluated eight different deep learning models consisting of ResNet, DenseNet, and the Inception network family, all of which achieved average F1 scores above 0.832, and the highest average value of 0.845 was obtained using InceptionResNetV2. Furthermore, incorporating the UQ demonstrated significant improvement in the overall model performance. Upon evaluation of the uncertainty estimate, MCD outperforms the other UQ models except for the metric, U_{Recall} , where DE and EMCD excel, implying that they are better at identifying incorrect predictions with higher uncertainty levels, which is vital in the medical field. Finally, we show that using a threshold for data referral can greatly improve the performance further, increasing the accuracy up to 0.959.

1. Introduction

According to the American Cancer Society, in 2023, an estimated 127,070 people (67,160 men and 59,910 women) will die from lung cancer, making the disease the leading cause of cancer death [1]. The likelihood of developing lung cancer is influenced by numerous factors, including tobacco smoking, the ageing process, family medical history, and exposure to environmental factors. Unfortunately, a significant number of cases result in a fatal outcome because they are diagnosed at an advanced stage when treatment is less effective. However, during the initial stages of this cancer, individuals may experience symptoms ranging from persistent coughing and chest pain to weight loss, which can sometimes be mistaken for other medical conditions. Therefore, early detection is of paramount importance. Screening with low-dose computed tomography (LDCT) has been demonstrated to be an effective method for detecting cancer while minimising exposure to harmful radiation [2]. Based on the Surveillance, Epidemiology, and End Results (SEER) database, if cancer is detected early and is localised only in the

part of the body where it originated, the 5-year relative survival rate can be as high as 62.8% [3]. An indicator of this cancer during screening is the presence of lung nodules, also known as pulmonary nodules. Most of them are usually benign, rarely malignant [4]. They can be measured up to 3 cm in diameter and have a rounded opacity that is almost well-defined [5]. Furthermore, these nodules can be categorised into three types; solid, part-solid, and ground-glass, and malignancy can be diagnosed based on size, content, shape, and colour [6].

Over the years, researchers have implemented numerous models using approaches ranging from traditional machine learning to deep learning to improve lung diagnosis. This involves nodule detection, segmentation, and classification. Because the presence of nodules does not always indicate cancer, the ability to differentiate between benign and malignant nodules is essential for extending patient survival. Different model architectures have been used, cited, and mentioned performance metrics such as accuracy and sensitivity of more than 90% [7–10]. Despite such numbers, challenges remain. One notable challenge is

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the model's reliability and robustness of its predictions, as highlighted in Wang [11]. Most of the existing models are deterministic, which means that they produce the same predictions with the same inputs. They perform well in classification, however, they might be overconfident if a new class is present, other than the existing labels. Here, uncertainty plays a major role. As models are usually seen as a 'black box', the models need to be able to say "*I don't know*", especially in dealing with medical data. Furthermore, models are typically evaluated on the basis of their performance with a specific dataset, making it challenging to determine whether the test data shares the same distribution, effectively presenting an out-of-distribution (OoD) problem. In addition, the cost of lung cancer treatment varies significantly, ranging from \$1,000 for surgery to \$4,809 for chemotherapy, as mentioned in Sheehan et al. [12]. This financial burden can be overwhelming for many families. Thus, with the control of uncertainty, extra information will be available. For instance, the certainty level associated with the prediction can help answer the question, "*Now that I know, what should I do?*" This, in turn, allows for personalised triaging of patients for tailored treatment plans.

Uncertainty by definition is the lack of confidence or knowledge about something. Instead of just a single prediction, a variable can be estimated by a distribution. In the context of deep learning, uncertainty arises due to randomness or noise in the data (aleatoric uncertainty) and uncertainty in the parameters of the model (epistemic uncertainty) [13]. These uncertainties can be modelled in various ways and have shown their benefit in medical applications [14]. In this regard, uncertain samples are referred to medical experts. Another application of uncertainty is in safety measures and adversarial attacks, in which a database shift or any perturbation to the input might cause overconfident predictions [15]. Furthermore, as highlighted by Gu et al. [7], there is a need to quantify uncertainty, especially in the field of lung cancer.

In this study, we investigate and explore the application of uncertainty quantification (UQ) techniques, specifically relying on Monte Carlo Dropout (MCD) and Deep Ensemble (DE), and employ them in the classification of lung cancer as either benign or malignant. We adopted a methodology inspired by Chagas et al. [16], beginning with the evaluation and selection of the most robust model, which is determined by the highest F1 score. Subsequently, we introduce a dropout layer to assess and incorporate uncertainty into our predictions.

The main contributions of our work are as follows:-

- We comprehensively explore a range of deep learning architectures and achieve favourable outcomes across this diversity,
- We employ three UQ techniques — Monte Carlo Dropout (MCD), Deep Ensemble (DE), and Ensemble Monte Carlo Dropout (EMCD) to quantify the model uncertainty by leveraging the best model with the highest F1 score as the base model,
- We conduct an in-depth investigation of the proposed UQ methods, exploring different dropout rates, and evaluating the impact of using an uncertainty threshold,
- We incorporate quantitative performance metrics to evaluate the quality of the uncertainty estimates, and
- We provide an extensive set of visualisations to enhance our understanding of how our models make predictions.

The remainder of this paper is structured as follows: In Section 2, we begin by discussing related works, starting with recent developments in lung cancer classification, followed by a review of uncertainty quantification and its applications in the medical field. Section 3 provides an explanation of the materials and methods employed, including data preparation, the model framework, and the techniques used for uncertainty quantification and measurement. Section 4 presents the experimental results and their discussion. Finally, Section 5 concludes the study and provides suggestions for future research directions.

2. Related works

2.1. Lung cancer classification

Diagnosing pulmonary nodules as cancer relies on the capability to analyse their internal structures and shapes. However, distinguishing between benign and malignant nodules is a challenging task. One of the primary reasons for this challenge is the nature of nodule characteristics, which results in interclass ambiguity (similar features but different classes) and intraclass variations (different features but the same class) [17]. Nonetheless, many efforts have been made to address this problem by proposing different model architectures, e.g., using 3D convolutional neural networks (CNNs), which can capture both spatial and temporal dimensions of the input data.

Dey et al. [18] compared four 3D neural networks using CNNs to classify pulmonary nodules as benign or malignant. Starting with an implementation of a basic 3D model, the author then utilised DenseNet and the idea of multi-pathway, which helped to obtain the highest accuracy and area under the curves (AUC) of 90.40% and 0.9548 respectively. In another study, Astaraki et al. [19] used the same idea of pathway and experiments with four deep learning models; VGG, ResNet, DenseNet, and EfficientNet. The reason behind the implementation is the strong correlation between intra-nodule heterogeneity and context information with nodule malignancy. Both supervised and unsupervised learning approaches were investigated, and a random forest model was used to generate the final classification results. The proposed model achieved an AUC of 0.93.

Moreover, Huang et al. [20] presented a two-stream network using DenseNet121 as the base model for their classification tasks. Due to the curse of dimensionality caused by deep features, a deep feature optimisation framework was proposed in this study to further enhance the performance. An accuracy of 92.13% and an AUC of 95.54% were obtained from the external validation dataset. In contrast, Xie et al. [21] commented on the lack of extensive training data sets, especially in the field of lung nodule classification. Therefore, the author used multiple fixed views of a 3D nodule and proposed a knowledge-based collaborative deep model. Multi-appearance submodels were also extracted and fused with the former views to improve classification accuracy. The model contains three pre-trained ResNet50 and outperforms existing approaches with an accuracy of 91.60% and an AUC of 95.70%.

In contrast to using multi-pathway and view, Xiao et al. [22] propose ensemble learning which includes three approaches for feature extraction and classification; supervised learning through ResNet18, unsupervised learning using an autoencoder, and classification by hand-craft features using an k-Nearest neighbour. To improve the generalisation ability, the results were combined and a weighted voting method based on the classification error rate was used. The overall performance achieved an accuracy of 93.10%. In addition, Muzammil et al. [23] presented ensemble learning through fusion techniques; averaging of prediction score, deep feature fusion, and maximum voting score. For feature extraction, AlexNet, VGG16, and VGG19 architectures were used, and three machine learning algorithms, SVM, LDA, and AdaBoostM2, were used for the classification phase. The best achieved accuracy was 96.89% and the AUC was 99.21%.

2.2. Uncertainty quantification

The need to know how confident a model is critical as being overconfident can also drive poor decisions. However, with the existence of uncertainty, complex samples can be identified and then referred for further investigation. A review of the recent advancements in UQ was presented in Abdar et al. [14], which covered different approaches and discussed various applications where it has been implemented and shown improvement to the existing system. Applications range from computer vision and medical applications to natural language processing. Moreover, in Gawlikowski et al. [24], factors that cause

uncertainty are highlighted, including variability in real-world situations, errors in measurement systems, model structure, training data, and unknown data. Generally, uncertainty can be categorised into two: epistemic uncertainty and aleatoric uncertainty [25]. Epistemic uncertainty is related to model uncertainty because of the lack of data to train the model. This can be reduced by introducing more data. On the other hand, aleatoric uncertainty is due to the noise of the data, and this type of uncertainty cannot be reduced.

To model such uncertainties approaches such as the Bayesian method or the non-Bayesian methods like the single deterministic method and the ensemble method are used. Based on the literature review, the first one was found to be the most popular. The Bayesian approach is based on the Bayes Theorem [26]. However, since the posterior distribution is intractable, approximations have been used instead; either sampling-based, which samples from a distribution of interest, or variational inference, an optimisation-based technique. As an example, in Combalia et al. [27], the initial approximation they employ is the Monte Carlo (MC) approach for modelling epistemic uncertainty. To estimate the aleatoric uncertainty, the authors have chosen to use test data augmentation. This technique generates multiple augmented examples from the same input, thereby yielding multiple predictions that can be leveraged for uncertainty estimation.

2.3. Uncertainty quantification in medical image classification

In this subsection, we have reviewed several existing studies that have employed UQ within the medical domain. Zahari et al. [28] proposed an uncertainty-aware framework to quantify uncertainty in 3D lung cancer image classification. They accomplished this using MCD and Test Time Augmentation (TTA) techniques, achieving the highest performance with an F1 score of 83.0%. Their approach involved the integration of a dropout layer (placed just before the classification layer) with a dropout rate probability of $p_{dropout} = 0.5$. The authors adopted the ResNet50 architecture and conducted a series of experiments, exploring configurations with and without class weights. They also applied the focal loss function, which yielded promising results. The study concludes that employing thresholds to identify uncertain images can significantly enhance model performance.

In the field of lung-related diseases, Ghoshal and Tucker [29] investigated the use of Dropweights-based Bayesian methods to estimate the uncertainty for the classification of COVID-19 disease. Two uncertainty measures were used; entropy and Bayesian Active Learning by Disagreement (BALD). Using a pre-trained ResNetV2, the author added Dropweights values of 0.1, 0.3, and 0.5 to the newly added fully connected layers on top of the backbone architecture to make it Bayesian and sampled the model with MC samples of 10, 25, and 50 times. Through their experimentation, the model accuracy was shown to have a strong correlation with the model uncertainty, and the model performance was 90% when 40% of the most uncertain samples were rejected. Moreover, Gour and Jain [30] incorporated an uncertainty-aware CNN for COVID-19 classification problems. Similar to the previous paper, the author changed the original top layer of EfficientNet-B3 but instilled just one dropout before the softmax layer and found that $p_{dropout} = 0.5$ was the optimal dropout value. The model was evaluated with MC samples 200 times, with the activation of dropout during the testing phase. The proposed model achieved a G-mean of 99.1% and a sensitivity of 99.30% for the binary classification problem. In this study, the confidence interval, standard deviation, and entropy were used as uncertainty measures.

On the other hand, UQs have also been incorporated into other medical applications. For example, in skin disease, Mobiny et al. [31] presented a Bayesian deep learning approach to improve the performance of skin lesion diagnosis. VGG16, ResNet50, and DenseNet169 models were assessed in their implementation. To make it Bayesian, the author has experimented with different locations for the dropout layer for the model and found that the DenseNet169 model with a dropout

layer applied after every dense block with $p_{dropout} = 0.5$ performs the best. By rejecting only 35% of the uncertain samples, the model performance reaches 90%. The study also concludes that model uncertainty is generally higher for incorrect predictions. Moreover, Abdar et al. [32] proposed a novel hybrid model for skin cancer classification using three UQ approaches (MCD, EMCD, and DE) through three-way decision. The author has implemented various deep learning models ranging from DenseNet and ResNet to MobileNet and has also employed a Bayesian optimisation approach to find the hyperparameter for fine-tuning the models. To assess the model performance, uncertain samples were obtained from the highest entropy of 50% and 20% of the data. While considering two sets of datasets, the final phase of the decision model attained the best accuracy, F1 score, and AUC of 90.96%, 91.00%, and 0.97, respectively. In addition, UQ is also used for kidney disease, where Chagas et al. [16] proposed an MCD approach for the classification of membranous nephropathy diseases using an evaluation pipeline that includes architecture evaluation as well as uncertainty estimation. The study experimented with $p_{dropout} = [0.3, 0.4, 0.5, 0.6, 0.7]$ and MC samples of $M = 10, 50, 100$ and considered two methods to quantify the uncertainty, MCD and TTA. Using $p_{dropout} = 0.4$, Wide-ResNet obtained the best F1 score of 0.935. Moreover, the author has demonstrated a relationship between variance as an uncertainty metric and the accuracy of predictions. In addition, this study includes a comparative analysis that considers five pathologists' judgments as a measure of complexity and compares them with the classification of uncertain images. However, no significant correlation was observed, suggesting the need for further experiments in this regard.

3. Materials and methods

In this experiment, we considered the Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI) dataset [33], or LIDC-IDRI in short, which is a collection of 1018 CT scans of lungs from various institutions. Four experienced radiologists have annotated the data and produced independent diagnostic results stored in an XML (Extensible Markup Language) file which includes a description of the nodules, degree of malignancy, and the eight semantic features (calcification, internal structure, lobulation, margin, sphericity, speculation, subtlety, and texture). For the pre-processing of data, we followed the steps are done in Zhao et al. [34], in which CT scans with slice thickness greater than 3 mm were excluded along with incomplete slices and inconsistent spacing. Furthermore, only nodules annotated by at least three of the four radiologists and with a diameter greater than or equal to 3 mm were considered in this study. For the ground truth, we computed the average malignancy score as our overall degree of malignancy. An average score below 3 is considered benign nodules, whereas scores above 3 are considered malignant. Nodules with an average of 3 were removed. This has also been done in other studies [35,36]. In addition, to obtain the best representation of the data, we performed anomaly detection to remove any outlier using density-based spatial clustering of applications with noise (DBSCAN) proposed by Ester et al. [37], and some nodules were discarded. The overall pre-process results in 1013 nodules consisting of 561 benign (55%) and 452 malignant (45%) nodules were used for the experiments.

We then extracted 3D images of $40 \times 40 \times 40$ dimensions based on the nodule centre coordination provided by the annotation. Moreover, we rescaled them to a fixed size of $75 \times 75 \times 75$ and resampled them with a uniform spacing of 1.0 mm^3 along the dimension using spline interpolation. Finally, to standardise the data, we used the min-max method to normalise the pixel intensity value measured in Hounsfield Unit (HU) scales with values of $(-1000, 400 \text{ HU})$ to a range of $(0, 1)$ [38]. Examples of benign and malignant nodules are shown in Fig. 1. To improve model generalisation and reduce overfitting, we implemented a real-time data augmentation with the help of 3D data augmentations [39]. Such transformations include random cropping and resizing, random rotation of 90 degrees, random flipping, random gamma, and random brightness and contrast, where 50% can occur for any training image given.

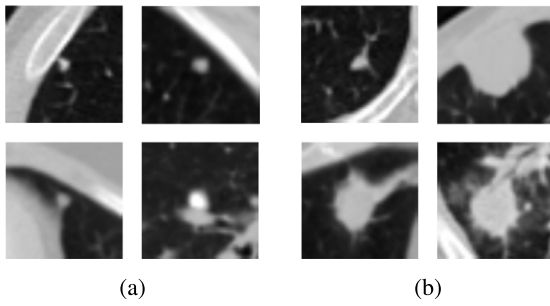


Fig. 1. Mid-slice examples from lung cancer categorised based on the average malignancy score as follows: (a) benign nodules (below level 3) and (b) malignant nodules (above level 3).

3.1. Proposed model framework

To provide an end-to-end solution, we have proposed an uncertainty-aware framework for the classification of benign and malignant nodules. Simultaneously, we can quantify model uncertainty, whereby uncertain images can be referred to medical experts upon selection of a threshold. Following the pipeline done in Chagas et al. [16], we have applied three UQ techniques instead of just one, as implemented in Abdar et al. [32], to analyse the robustness of the models. The UQ methods are MCD, DE, and a combination of the two, which is EMCD. The pipeline consists of three phases, as shown in Fig. 2. The first phase deals with the preprocessing of input data, which involves resizing, resampling, and anomaly detection. Once the data represented the best distribution, we split them into training, validation, and testing data and fed them for model selection and evaluation. The best model with the highest F1 score will be chosen as the base model.

Moving on to the next phase, which is quantifying the uncertainty, for the MC approach, we decided to include a dropout layer just before the classification layer to make it Bayesian and thus, probabilistic. We retrain the model by considering different $p_{dropout}$ and M (number of forward passes, or MC samples) values. By running multiple passes, multiple predictions can be obtained, thus estimating the predictive distribution. However, as for the DE approach, we combine the prediction from the three base models for the final prediction.

For the third and final phase, these predictions are categorised as correct and incorrect. We used entropy and standard deviation as the uncertainty measurement, and by varying this measurement as a threshold, uncertain images can then be referred to medical experts, and only the retained data are computed to measure the performance for the final predictions. In addition, we quantitatively evaluate the quality of the uncertainty estimates.

3.2. Model architecture selection

As mentioned earlier, the first phase of the proposed framework deals with the selection of models. Below we briefly describe the architecture used in this study:

- **ResNet** [40]: Utilises skip connections to overcome the vanishing gradient problem, allowing for deeper networks.
- **DenseNet** [41]: Features densely connected blocks to enhance feature reuse and reduce the vanishing gradient issue.
- **InceptionResNet** [42]: Combines Inception's multi-scale feature learning with ResNet's residual connections, facilitating deeper networks.
- **Inception** [43]: Employs Inception modules for capturing features at various scales, leading to a wider, rather than deeper, network structure.

For this study, we implemented a total of eight models; ResNet18, ResNet34, ResNet50, ResNet101, DenseNet121, DenseNet169, InceptionResNetV2, and InceptionV3. Upon evaluation, the model with the highest F1 score will be selected as the UQ base model.

3.3. Uncertainty quantification

To quantify or estimate the uncertainty, we applied three UQ techniques and provided a brief explanation of how they work.

3.3.1. Monte Carlo Dropout (MCD)

MCD is a popular method for uncertainty quantification. As proposed by Gal and Ghahramani [44], a predictive distribution can be learned in a scalable way by leveraging dropout regularisation. Dropouts are utilised as a means to prevent overfitting. It works by randomly dropping or removing the neurons, which then prevents the memorisation of the training data. Generally, it is used only at training time, however, MCD extends this mechanism by using them both during training and testing time and helps to produce predictive distribution by evaluating the input many times and running several forward passes of the model. The softmax outputs for each class can then be averaged to obtain the final prediction and to express the model's uncertainty.

Consider a trained network model, f^{w^*} where f is the model network and w^* is the model weights parameters with an active dropout. Suppose a new test $\{x^*, y^*\}$ is to be predicted, where x is the image array and y is the associated label. It will be passed through the network with active dropout M times (i.e. no. of forward passes). Thus, the predictive posterior distribution can be approximated to generate the final prediction as follows:

$$p(y^*|x^*) = \frac{1}{M} \sum_{m=1}^M p_m^*, \quad (1)$$

where p_m^* denotes the output of the softmax function at pass $m \in \{1, \dots, M\}$ and defined by Eq. (2).

$$p_m^* = \text{Softmax}(f^{w_m^*}(x^*)). \quad (2)$$

3.3.2. Deep Ensemble (DE)

An alternative to the Bayesian network, a simple and scalable approach was introduced by Lakshminarayanan et al. [45], which has been found to capture more reliable uncertainty. This approach estimates model uncertainty by taking advantage of an ensemble of models. Training multiple models with different sets of parameters produces several possible predictions and their corresponding uncertainties. A probability distribution over the model's parameter can then be placed and used to estimate the likelihood of different predictions. Although it outperforms MCD, it can be computationally expensive to perform model training and testing.

3.3.3. Ensemble Monte Carlo Dropout (EMCD)

The EMCD approach leverages a combination of models not only with simple deterministic but also equipped with MCD. This means that every model in the ensemble can quantify uncertainty individually by enabling dropout during both the training and testing phases. Following MCD, during the testing phase, each model in the ensemble undergoes multiple forward passes with dropout activated. This process generates a set of predictions for each input sample, reflecting the distribution of possible outcomes due to the randomness introduced by dropout. The core of this approach is the aggregation of these diverse predictions across all models and their respective MC simulations. For every input sample, the predicted probability for each class is averaged, resulting in the final prediction, which is used to express the uncertainty. By leveraging multiple models, the risk of overfitting and biases is reduced. Simultaneously, through MCD's random dropout, each model's predictions capture a range of possible outcomes.

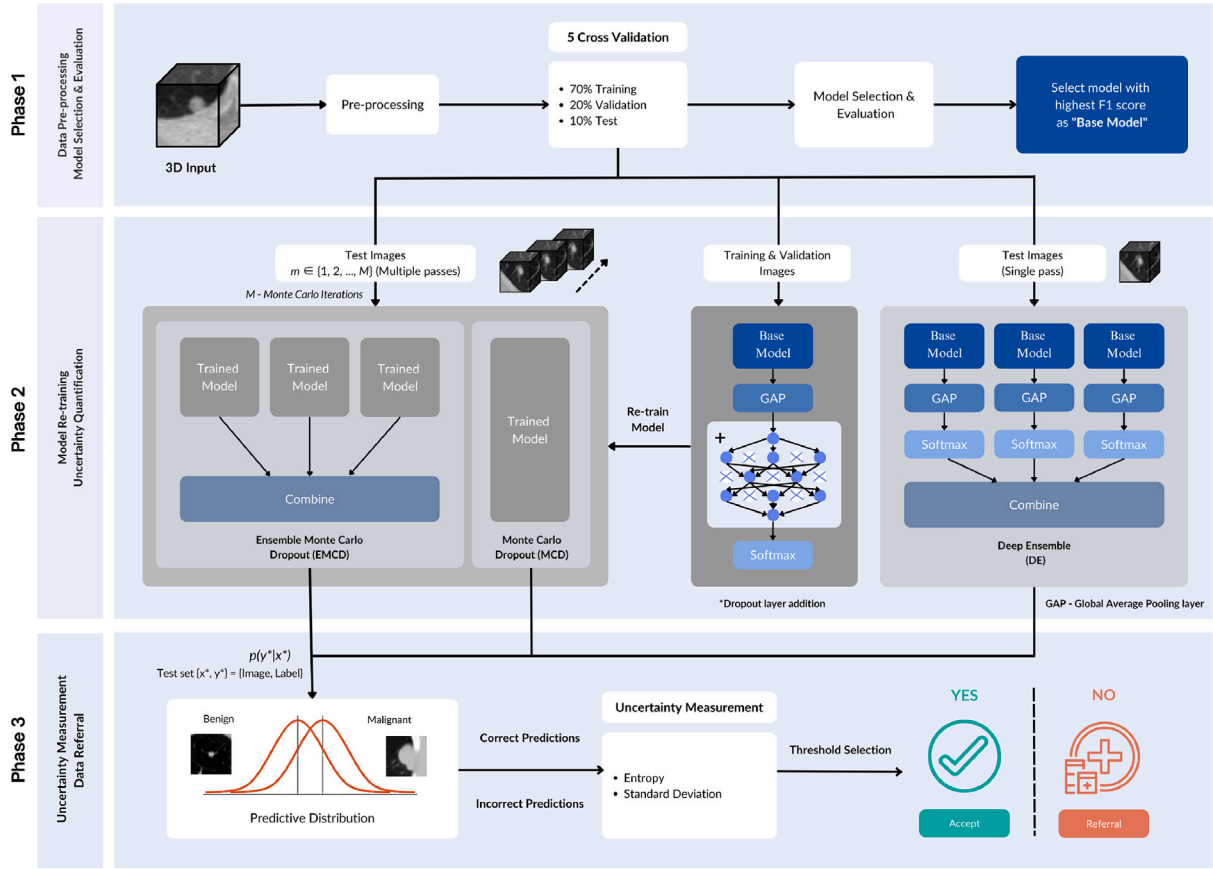


Fig. 2. An overview of the proposed framework for quantifying uncertainty in the classification of lung nodules as benign or malignant. This framework consists of three phases: beginning with model selection and evaluation, followed by uncertainty quantification, and concluding with uncertainty measurement and data referral. The input data comprised 3D lung nodule images, which were preprocessed and split into training, validation, and testing sets before model training. The base model, selected based on the highest F1 score, incorporates a newly introduced dropout layer just before the final classification layer. Three UQ techniques were used to estimate the uncertainty and then entropy and standard deviation as the uncertainty measurement. Model performance is assessed based on a specified threshold, considering the standard performance metrics and using quantitative uncertainty evaluation metrics to evaluate the quality of the uncertainty estimates.

3.4. Uncertainty measurement

To measure model uncertainty, we use two different metrics: entropy and standard deviation.

- Entropy [46]: A measure of variability associated with random variables and is defined by the following equation.

$$H(p(y^*|x^*)) = - \sum_{c=1}^C p(y^*|x^*) \log p(y^*|x^*), \quad (3)$$

where C is the number of classes.

- Standard Deviation: A measure of how dispersed the data are to the mean. It can be defined by the following equation.

$$\sigma = \sqrt{\frac{1}{M} \sum_{m=1}^M [p_m^* - p(y^*|x^*)]^2}. \quad (4)$$

A higher measurement implies greater uncertainty. We also decided to normalise the uncertainty values, H_{norm} and σ_{norm} again using the min-max method in the range of 0 to 1, which will enable us to make comparisons across different configurations. The decision to use these metrics is based on the theoretical basis and widespread validation in the literature as effective tools for quantifying uncertainty in deep learning models [29,32,47].

Moreover, we computed an uncertainty ratio, which is defined as the relationship between the average entropy of incorrect predictions and that of correct predictions. This additional metric then enables us to assess the correlation between entropy and the accuracy of predictions.

3.5. Uncertainty assessment

To assess a generic machine learning model, a confusion matrix is used, which summarises the performance on a set of test data. It is a visual representation between the actual and predicted values where various metrics can be derived from this matrix, for instance, accuracy. With recent advances in the ability to quantify uncertainty, additional information is available to improve the decision process. However, it remains a hot topic to assess uncertainty values [48]. Nonetheless, Asgharnezhad et al. [49] proposed a novel approach to quantitatively evaluate uncertainty estimates by introducing an uncertainty-aware confusion matrix. It is similar to the general matrix but requires an additional metric; uncertainty value. The actual or ground truth is compared with the predicted truth and then categorised into correct and incorrect predictions. With uncertainty, a selection of thresholds is applied to further split the predictions into certain and uncertain. Thus, the uncertainty confusion matrix has four outcomes (TC, TU, FC and FU), as shown in Fig. 3 where the desired outcomes are coloured green and the undesired outcomes are coloured red.

Below is a brief explanation of the outcomes:-

- True Certainty (TC): number of correct predictions and is certain about it,
- False Certainty (FC): number of incorrect predictions and is certain about it,
- True Uncertainty (TU): number of incorrect predictions and is uncertain about it, and

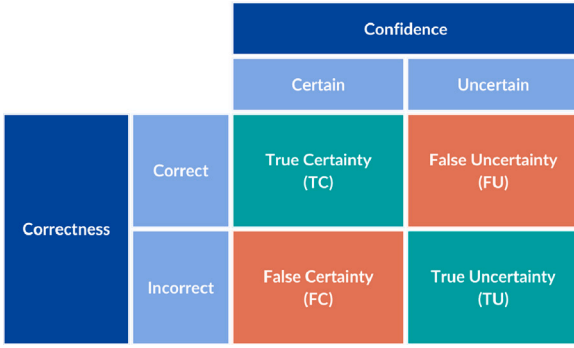


Fig. 3. Uncertainty confusion matrix. The obtained predictions are compared with the ground truth and then placed in either correct or incorrect predictions. Next, using a predetermined threshold, these predictions are further classified into certain and uncertain.

- False Uncertainty (FU): number of correct predictions and is uncertain about it

The certainty of the predictions is based on the threshold, which means that different values of a chosen threshold produce different sets of matrices. From Fig. 3, analogous to the conventional confusion matrix, performance metrics can be further defined:-

- $U_{Accuracy}$ is how often the model makes the right predictions, as in this case, will be the TC and TU. Thus, it can be calculated by dividing the diagonal outcomes by the total number of outcomes.

$$U_{Accuracy} = \frac{TU + TC}{TU + TC + FU + FC}. \quad (5)$$

- $U_{Precision}$ is the proportion of uncertain predictions that were incorrect. It is defined as the number of incorrect and uncertain predictions divided by the total number of uncertain predictions:

$$U_{Precision} = \frac{TU}{TU + FU}. \quad (6)$$

- U_{Recall} is the proportion of incorrect predictions identified as uncertain. It is defined as the number of incorrect and uncertain predictions divided by the total number of incorrect predictions:

$$U_{Recall} = \frac{TU}{TU + FC}. \quad (7)$$

- $U_{Specificity}$ indicates the proportion of correct predictions that were actually certain. It can be calculated by dividing the number of correct and certain predictions by the total number of correct predictions:

$$U_{Specificity} = \frac{TC}{TC + FU}. \quad (8)$$

An indication of when the model performs better is when these metrics have high values. All the performance metrics mentioned above are a function of their values, which means that their values will change with changes in the uncertainty threshold.

4. Results and discussions

4.1. Experimental setup

We provide further details on the experimental settings of the proposed model. We have followed the strategy protocol done in [30,50], where the dataset is split using a 5-fold cross-validation, in which the training, validation, and test set are in the ratio of 70:10:20. As the test set is across the entire dataset, rather than just one hold-out, a fair comparison can be made with the existing approaches. All

Table 1

Comparison of the results showing the mean and standard deviation provided by the selected models for classification of benign and malignant nodules from LIDC-IDRI datasets.

Model	μ Accuracy	μ AUC	μ Precision	μ Recall	μ F1 score
ResNet18	0.841 (± 0.028)	0.915 (± 0.025)	0.843 (± 0.029)	0.836 (± 0.029)	0.837 (± 0.029)
ResNet34	0.829 (± 0.032)	0.904 (± 0.022)	0.833 (± 0.033)	0.822 (± 0.032)	0.825 (± 0.033)
ResNet50	0.827 (± 0.028)	0.909 (± 0.021)	0.828 (± 0.030)	0.822 (± 0.029)	0.823 (± 0.029)
ResNet101	0.830 (± 0.032)	0.911 (± 0.023)	0.833 (± 0.034)	0.824 (± 0.033)	0.826 (± 0.033)
DenseNet121	0.835 (± 0.028)	0.909 (± 0.023)	0.837 (± 0.029)	0.830 (± 0.029)	0.832 (± 0.029)
DenseNet169	0.837 (± 0.026)	0.913 (± 0.026)	0.841 (± 0.029)	0.831 (± 0.026)	0.833 (± 0.026)
InceptionResNetV2	0.848 (± 0.029)	0.925 (± 0.023)	0.851 (± 0.032)	0.842 (± 0.028)	0.845 (± 0.029)
InceptionV3	0.839 (± 0.029)	0.917 (± 0.024)	0.843 (± 0.031)	0.835 (± 0.027)	0.836 (± 0.029)
	0.836 (± 0.029)	0.913 (± 0.024)	0.839 (± 0.031)	0.830 (± 0.029)	0.832 (± 0.029)

model architectures were modelled, trained, and evaluated using the TensorFlow framework [39]. We used the Adam optimiser to update the weights of the network, and the epoch was set to 500 with a batch size of 8. The learning rate in our study was set to 1×10^{-5} , and we incorporated early stopping mechanisms to prevent overfitting. To ensure the reliability of our results, we conducted each experiment three times, reporting both the mean and its associated standard deviation. These experiments were conducted on a machine equipped with an NVIDIA GeForce RTX 3070 GPU with 16 GB of memory. For the initial phase of evaluating the classification models, we employed the conventional metrics commonly used in existing studies, including accuracy, the area under the curve (AUC), precision, recall, and F1 score. In the second phase, which focused on assessing uncertainty estimates, we used quantitative uncertainty evaluation metrics.

4.2. Model evaluation

In this section, we present the results and findings from the first phase, which involved model selection and evaluation. We conducted experiments using various models, as described in Section 3.2, to determine which model performs best in terms of the highest F1 score. Table 1 provides the mean results and standard deviation for the models used in the classification of benign and malignant nodules. The last row summarises the overall performance of all models, and it is evident that all models achieve performance metrics above 0.832, with a relatively low standard deviation ranging from approximately 0.026 to 0.033.

The model with the highest F1 score, 0.845, was InceptionResNetV2, while the lowest performing model, ResNet34, achieved an F1 score of 0.825, representing a difference of 0.020. Importantly, the low standard deviation values indicate consistent and stable results across different cross-validation folds. The model with the lowest standard deviation, DenseNet169, achieves a value of 0.026, although it is not significantly different from the other models. Our study did not reveal a clear correlation between the depth of ResNet architecture and performance, as ResNet18 outperformed the deeper variants. Regarding the DenseNet family, DenseNet169 demonstrated better performance than DenseNet121. However, further experiments may be necessary to determine whether denser networks offer improved performance. Unfortunately, the expensive computational requirements prevented us from implementing DenseNet201, which has 201 layers.

Another important consideration when selecting a model is to evaluate its performance efficiency, which includes factors such as model size and capacity. Table 2 provides information on the number of trainable parameters for each model architecture. The parameter count

Table 2
Number of training parameters in millions(M) for the selected model architectures.

Model	# trainable parameters
ResNet18	33.163 M
ResNet34	63.473 M
ResNet50	46.152 M
ResNet101	85.198 M
InceptionResNetV2	67.452 M
InceptionV3	34.085 M
DenseNet121	11.245 M
DenseNet169	18.546 M

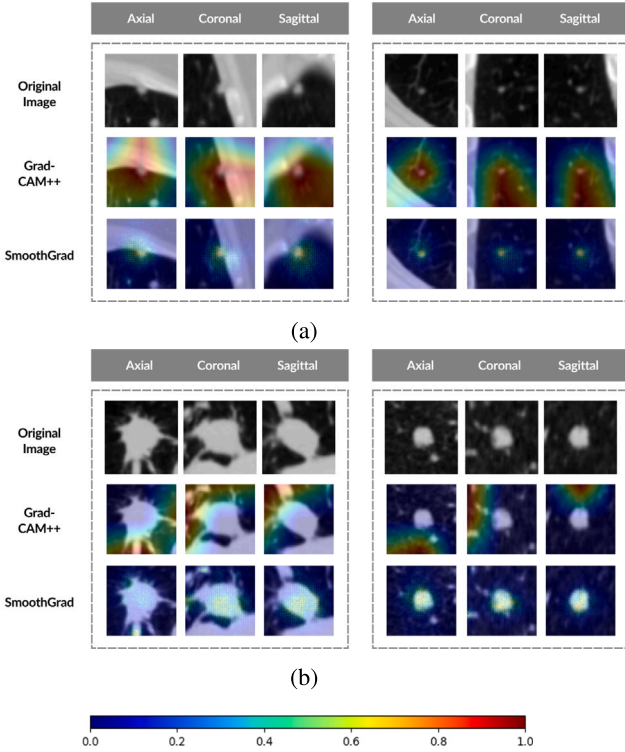


Fig. 4. GradCAM++ and SmoothGrad visualisation for two samples (left and right) for (a) benign nodules and (b) malignant nodules with views from three different planes; axial, coronal and sagittal.

varied from the smallest model, DenseNet121, with 11.245 million parameters, to the largest model, ResNet101, with 85.198 million parameters. An alternative model choice worth considering is ResNet18, which has approximately half the number of parameters compared with InceptionResNetV2 and achieved the second-best performance in the model evaluation. This represents a favourable trade-off between performance and efficiency. ResNet18 is not only quicker to train and test but also demands fewer memory resources. In addition, it is worth noting that the DenseNet family of models has the fewest parameters compared with the other architectures. Despite their smaller size, they perform admirably in classifying nodules compared with models from the ResNet family.

Moreover, to gain insights into how and where the model makes its class predictions, we employed two visualisation approaches: Grad-CAM++ [51] and SmoothGrad [52]. The outcomes of these visualisations are presented in Fig. 4, which shows (a) benign nodules and (b) malignant nodules from various perspectives, including axial, coronal, and sagittal views. In these visualisations, the intensity of the colours on the heatmap corresponds to the image features that are considered significant for lung nodule classification.

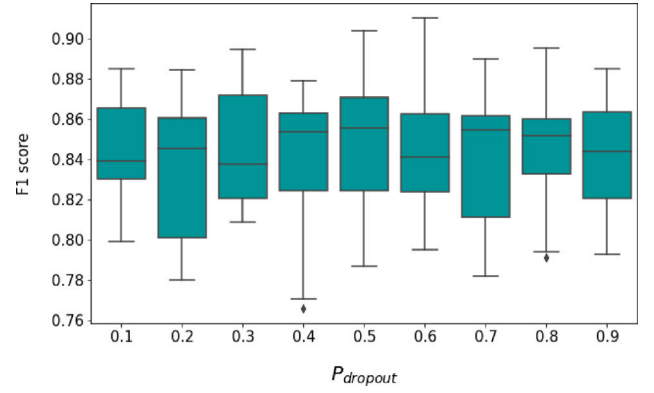


Fig. 5. Box and whisker plot of the F1 score for the variation of dropout rates, $P_{dropout}$ ranging from 0.1 to 0.9 with a stepsize of 0.1 used in the classification problem. On each box, the central line marks the median, and the edges of the box are the lower quartile (25th percentile) and upper quartile (75th percentile). The small dots represent outliers. The higher the F1 score, the better the performance.

Based on the Grad-CAM++ visualisation, for benign nodules, the highlighted overlays cover larger, more diffuse areas around the nodule. This suggests that the model's focus is distributed across the nodule and its surrounding tissue without targeting any specific aggressive features. In contrast, for malignant nodules, the overlays show intense and localised hotspots, particularly at defined edges or irregular shapes. This indicates that the model has pinpointed specific regions of interest that could indicate malignancy.

SmoothGrad visualisations of benign nodules indicate a distinct, localised area of focus within the nodules. Contrary to the expected uniformity, the model identified specific features within the nodules that were influential in the model's assessment. For malignant nodules, SmoothGrad overlays also exhibit bright, concentrated spots. These spots are more pronounced, reflecting the model's focus on irregularities or areas of concern that align with the typical features of malignancy. It is crucial to recognise that these interpretations are heavily dependent on the accuracy and training of the deep learning model. Hence, while such visualisations are intended to be assistive tools, further validation by medical experts is essential.

4.3. Uncertainty quantification

We then assess the performance of the proposed models: MCD, DE, and EMCD. For the best-performing model, InceptionResNetV2, a dropout layer is introduced just before the classification layer. To estimate the MCD, we conducted experiments with varying dropout rates, $P_{dropout} \in [0.1 - 0.9]$, using a step size of 0.1. The model was retrained using different configurations and parameters. The results based solely on the F1 score and the variation in the dropout rates are presented in Fig. 5. It is evident from the results that the F1 score remains relatively consistent across all models, with only a slight variation. The mean F1 score ranged from the lowest value of 0.833 to the highest value of 0.847. Our experimentation indicates that using $P_{dropout} = 0.5$ achieved the best performance in terms of achieving the highest F1 score. This F1 score is slightly higher than the value obtained from the model without uncertainty, which is 0.845, as shown in Table 1.

In the case of DE, as the models undergo three separate training processes, predictions from the three InceptionResNetV2 models are combined, and the final prediction is determined by averaging their outputs. The variances among these model predictions can then be used to calculate the uncertainty. It is important to note that the size of the ensemble significantly impacts the model's performance. Larger ensembles may offer enhanced predictive capabilities and improved accuracy [53]. However, the effectiveness of varying ensemble sizes

Table 3

Comparison of the obtained results with their mean and standard deviation of the three types of UQ methods using InceptionResNetV2 as the base model for the LIDC-IDRI dataset. In this table, Entropy-Correct (ET-Co) indicates the entropy of correctly classified data, Entropy-Incorrect (ET-Inc) indicates the entropy of incorrectly classified data, Standard Deviation-Correct (STD-Co) indicates the standard deviation of correctly classified data, Standard Deviation-Incorrect (STD-Inc) indicates the standard deviation of incorrectly classified data, and Unc. The ratio is the uncertainty ratio based on the entropy.

Model	μ Accuracy	μ AUC	μ Precision	μ Recall	μ F1 score	μ ET-Co	μ ET-Inc	μ STD-Co	μ STD-Inc	μ Unc. Ratio
MCD	0.850 (± 0.030)	0.925 (± 0.022)	0.853 (± 0.029)	0.844 (± 0.032)	0.847 (± 0.031)	0.301 (± 0.059)	0.655 (± 0.077)	0.204 (± 0.047)	0.416 (± 0.113)	2.234 (± 0.348)
DE	0.860 (± 0.019)	0.933 (± 0.022)	0.863 (± 0.023)	0.855 (± 0.016)	0.857 (± 0.018)	0.342 (± 0.044)	0.725 (± 0.054)	0.154 (± 0.025)	0.358 (± 0.026)	2.143 (± 0.249)
EMCD	0.869 (± 0.030)	0.938 (± 0.018)	0.872 (± 0.027)	0.863 (± 0.033)	0.866 (± 0.032)	0.358 (± 0.036)	0.746 (± 0.029)	0.170 (± 0.011)	0.416 (± 0.056)	2.102 (± 0.253)

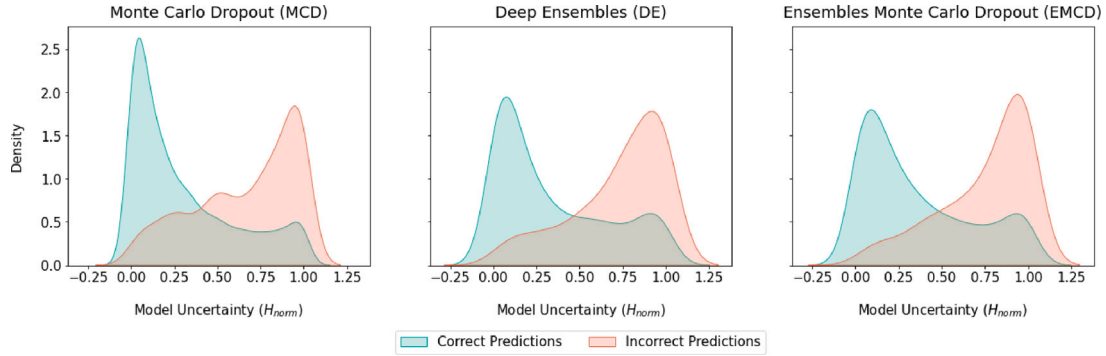


Fig. 6. Distribution of normalised model uncertainty values, H_{norm} for the three UQ methods, based on the test data and categorised between correct (green) and incorrect (red) predictions. This shows that incorrect predictions have higher model uncertainty, prompting uncertain images to be referred to medical experts and thus improving the performance of the model. The density (vertical axis) representing the output posterior distributions is based on Gaussian kernel density estimation.

is also highly dependent on the specific task at hand, the dataset, and the availability of computational resources. Thus, the choice of a three-model ensemble represents a starting and initial step, but such an approach has been substantiated in existing literature, particularly in studies related to UQ [32] and lung [54,55].

On the other hand, EMCD represents a combination of both methods. In this case, ensembles of InceptionResNetV2 models with a dropout rate of $p_{dropout} = 0.5$ are employed. The comprehensive results are presented in Table 3. The highest F1 score was achieved by the EMCD, amounting to 0.866. However, DE exhibited the smallest standard deviation with an F1 score of 0.857. It is worth noting that while MCD yields the highest uncertainty ratio, its standard deviation is comparatively higher than the others.

We conducted further analysis using the uncertainty values generated by the UQ models. In the experiment, we investigated whether a correlation exists between the uncertainty values and the model predictions, which warrants a more in-depth exploration. To visualise this, we plotted the model uncertainty for the test images across all folds. We employed the kernel density estimation with a Gaussian kernel to create a visual representation. The results were categorised based on correct and incorrect predictions, as illustrated in Fig. 6. Observations from the plot reveal that higher model uncertainty is associated with incorrect predictions across all three UQ models. This observation implies that selecting uncertainty values can serve as a means of identifying uncertain cases, mirroring the clinical workflow. This approach allows for the referral of such cases to medical experts for a secondary evaluation, potentially leading to an enhancement in the overall model performance.

To further demonstrate the robustness and reliability of the model, we also conducted experiments with two additional medical datasets: SynapseMNIST3D and VesselMNIST3D, obtained from MedMNIST [56]. SynapseMNIST3D consists of either excitatory or inhibitory synapses, with data acquired using a multi-beam scanning electron microscope. On the other hand, VesselMNIST3D comprises datasets of entire brain vessels, which were collected by reconstructing magnetic resonance angiography (MRA) images and labelled as either healthy or aneurysm

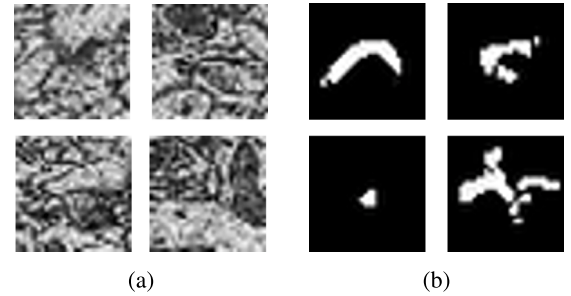


Fig. 7. Mid-slice examples from (a) SynapseMNIST3D, showing top: excitatory, and bottom: inhibitory synapses; and (b) VesselMNIST3D, showing top: healthy, and bottom: aneurysm segments.

segments. These datasets were standardised into a small 3D volume size of $28 \times 28 \times 28$. Examples of SynapseMNIST3D and VesselMNIST3D datasets are shown in Fig. 7.

For data preprocessing, we resized the volumes to $32 \times 32 \times 32$ to meet the minimum input requirements of the model architectures. Beyond this resizing, we assume that the data, as presented, best represents the underlying distribution suitable for the model. Subsequently, we performed the same experiment to identify the optimal model, which resulted in InceptionResNetV2 being selected for both datasets. As before, we incorporated $p_{dropout} = 0.5$ for both the MCD and EMCD techniques. For DE, we used three sets of models. The comprehensive results for both datasets are presented in Table 4. It was observed that DE consistently achieved the highest scores across both datasets, highlighting that ensembling multiple models typically results in superior performance. This is attributed to the diversity in ensemble learning, where a diverse set of models covers a broader range of possibilities and errors. The ensemble's ability to aggregate predictions means that individual errors are likely to cancel out or reduce, leading to more accurate and robust predictions [53].

Table 4

Comparison of the obtained results for validation with their mean and standard deviation of the three types of UQ methods using InceptionResNetV2 as the base model for (a) SynapseMNIST3D and (b) VesselMNIST3D datasets.

Dataset	Model	μ Accuracy	μ AUC	μ Precision	μ Recall	μ F1 score	μ ET-Co	μ ET-Inc	μ STD-Co	μ STD-Inc	μ Unc. Ratio
(a)	MCD	0.871	0.918	0.846	0.821	0.830	0.290	0.663	0.210	0.483	2.359
		(± 0.024)	(± 0.024)	(± 0.035)	(± 0.034)	(± 0.031)	(± 0.063)	(± 0.07)	(± 0.064)	(± 0.103)	(± 0.428)
	DE	0.898	0.942	0.881	0.853	0.865	0.354	0.769	0.189	0.448	2.190
		(± 0.018)	(± 0.014)	(± 0.023)	(± 0.027)	(± 0.024)	(± 0.024)	(± 0.059)	(± 0.018)	(± 0.072)	(± 0.308)
	EMCD	0.898	0.940	0.883	0.850	0.864	0.380	0.765	0.199	0.430	2.024
		(± 0.02)	(± 0.019)	(± 0.027)	(± 0.029)	(± 0.027)	(± 0.047)	(± 0.059)	(± 0.022)	(± 0.064)	(± 0.16)
(b)	MCD	0.963	0.969	0.931	0.877	0.900	0.094	0.585	0.067	0.405	7.205
		(± 0.012)	(± 0.014)	(± 0.035)	(± 0.042)	(± 0.034)	(± 0.038)	(± 0.114)	(± 0.031)	(± 0.149)	(± 3.572)
	DE	0.973	0.986	0.945	0.918	0.930	0.110	0.655	0.054	0.421	6.261
		(± 0.008)	(± 0.009)	(± 0.029)	(± 0.014)	(± 0.018)	(± 0.023)	(± 0.092)	(± 0.015)	(± 0.113)	(± 2.087)
	EMCD	0.968	0.983	0.945	0.891	0.915	0.122	0.691	0.064	0.423	6.256
		(± 0.011)	(± 0.003)	(± 0.036)	(± 0.026)	(± 0.029)	(± 0.043)	(± 0.089)	(± 0.022)	(± 0.089)	(± 2.415)

Conversely, MCD exhibited the lowest performance metrics. While dropout is a valuable technique for uncertainty estimation, it may not always enhance performance [57]. This can be attributed to the noise introduced by the variability in output predictions due to the stochastic nature of dropout during inference. As different neurons are dropped in each forward pass, the network's processing path changes, leading to variations in output. The theoretical foundation for the MCD approach comes from Bayesian neural networks, where dropout is used to approximate Bayesian inference, providing a practical and computationally efficient method to estimate model uncertainty [44]. However, its reliance on the dropout rate means that the quality of UQ can vary, highlighting the importance of careful model tuning and validation.

Furthermore, EMCD theoretically proposes a balance between the robust performance of DE and the subtlety uncertainty estimation of MCD. Its performance, while not always surpassing that of DE, provides competitive or improved uncertainty measures. This indicates that EMCD successfully leverages the strengths of both MCD and DE, offering a balanced approach with the potential for better UQ in ambiguous or complex scenarios. Moreover, the observed pattern in entropy and standard deviation for correct and incorrect predictions affirm that advanced UQ mechanisms enable models to more effectively differentiate between certain and uncertain predictions.

4.4. Uncertainty estimates evaluation

In the previous section, we analysed the overall performance of the UQ model. Notably, the inclusion of dropout led to performance enhancements, and when we further employed model ensembling, performance improved even more. However, in this section, our primary focus is on evaluating and assessing the quality and accuracy of uncertainty estimates. To accomplish this, we incorporated metrics derived from Section 3.5. Specifically, we opted for the uncertainty measure, entropy, H_{norm} , as our threshold value and recalculated the relevant metrics across a range of values ranging from 0.01 to 0.99 with a stepsize of 0.01. We chose these values deliberately to achieve precision in our results and to provide a clearer visualisation of how these metrics vary. It is essential to note that when the threshold value is set to 0, all predictions are uncertain, whereas a threshold value of 1 signifies complete certainty in the predictions.

Fig. 8 presents the results for all three UQ models, and it is evident that they follow the same pattern with no significant differences among them. Notably, metrics such as $U_{Accuracy}$, $U_{Precision}$, and $U_{Specificity}$ exhibit a positive correlation, whereas U_{Recall} demonstrates a negative correlation. This decreasing trend aligns with our expectations because it corresponds to an increase in incorrect predictions with rising threshold values, which contributes to false uncertainties (FU). Higher threshold values indicate better model performance. Upon observation, we found that all performance metrics approach or reach a value of 1

or come very close to it, except for $U_{Precision}$. This deviation can be attributed to the presence of correct predictions at higher threshold values.

We then analyse these metrics at high (> 0.5) and low (< 0.5) threshold values. At high threshold levels, $U_{Accuracy}$ is quite high for all methods, implying that the model's ability to detect TU and TC is better. This is expected as a higher threshold generally implies a stricter criterion for making the right predictions. $U_{Precision}$ and U_{Recall} show variability among the methods. $U_{Precision}$, which indicates the proportion of uncertain predictions that were incorrect, is moderately low across all methods, suggesting that when the model flags a prediction as uncertain, it is not always incorrect. This could imply an overly cautious uncertainty estimation. U_{Recall} , indicating the ability to capture incorrect predictions as uncertain, varies more significantly, with EMCD showing the highest mean, suggesting it is more effective at identifying incorrect predictions under high uncertainty thresholds. $U_{Specificity}$ is high across all methods, indicating that correct predictions are confidently identified as such. This is an essential characteristic of reliable models, as it ensures that the model is not only accurate but also confident in its correct predictions.

At a low threshold (< 0.5), $U_{Accuracy}$ drops significantly for all methods, suggesting that as the threshold for uncertainty is lowered, the model's overall correctness decreases. This is to be expected, as reducing the threshold leads to a higher number of predictions being classified as uncertain, including both correct and incorrect ones. $U_{Precision}$ remains low, which implies that many correct predictions are considered uncertain whereas U_{Recall} is very high for all methods at low thresholds, indicating a strong ability to flag incorrect predictions as uncertain. This is beneficial for applications where missing an incorrect prediction has high costs, as it allows for human intervention or further analysis of uncertain cases. $U_{Specificity}$ decreases compared to high thresholds, which is expected as more correct predictions are also considered uncertain at lower thresholds.

To conduct a more in-depth analysis of these metrics, we selected two specific thresholds, namely 0.50 and 0.75. This choice allowed us to compare the performance of the three UQ models, and the results are summarised in Tables 5a and b. It is clear that for all models, there is a noteworthy improvement compared with their initial performance for both thresholds. This underscores how the implementation of uncertainty models can effectively identify incorrect predictions. For both thresholds, it is worth noting that the MCD model stands out with the highest $U_{Accuracy}$, signifying that this model excels in identifying TC and TU while minimising the number of uncertain data referrals. This result is likely attributable to the model's ability to make more accurate predictions, as observed in the distribution of the uncertainty values illustrated in Fig. 6. However, it is important to highlight that the differences between the other models are not substantial. The same trend is observed when considering $U_{Specificity}$. Furthermore, the capability to flag incorrect predictions with high uncertainty holds significant importance, particularly in fields such as medicine, where uncertain

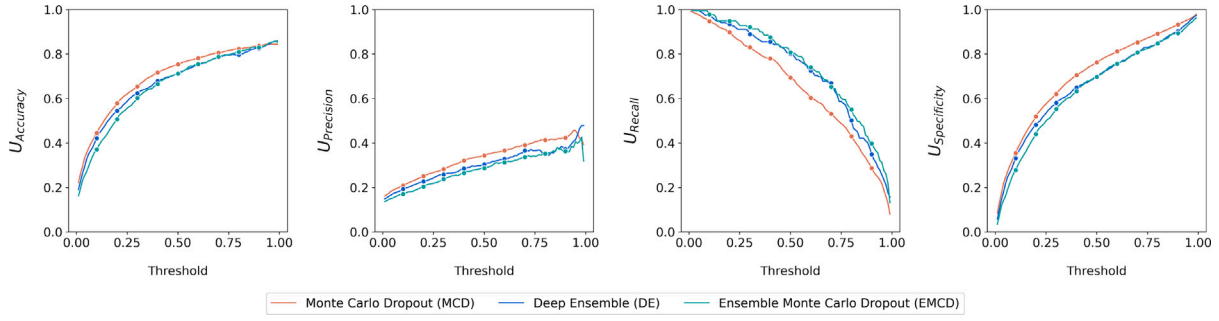


Fig. 8. Analysis of the quantitative uncertainty estimate performance showing the mean using the evaluation metrics; $U_{Accuracy}$, $U_{Precision}$, U_{Recall} , and $U_{Specificity}$. The metrics were calculated for varying thresholds of uncertainty between 0.01 and 0.99 with a stepsize of 0.01. Note that the rounded tick labels represent an interval of 0.1 starting from 0.1.

Table 5

The mean performance metrics along with the standard deviation for the specific threshold (a) 0.50 and (b) 0.75 for all the UQ models.

(a)			
Metrics	MCD	DE	EMCD
$\mu Accuracy$	0.934 (± 0.024)	0.955 (± 0.026)	0.959 (± 0.014)
$\mu F1$ score	0.932 (± 0.026)	0.954 (± 0.027)	0.958 (± 0.014)
$\mu U_{Accuracy}$	0.753 (± 0.051)	0.713 (± 0.044)	0.711 (± 0.036)
$\mu U_{Precision}$	0.343 (± 0.070)	0.303 (± 0.023)	0.286 (± 0.050)
μU_{Recall}	0.692 (± 0.105)	0.800 (± 0.114)	0.807 (± 0.037)
$\mu U_{Specificity}$	0.762 (± 0.064)	0.699 (± 0.063)	0.696 (± 0.043)
% Data Retained	69.422 (± 6.638)	62.800 (± 6.573)	63.000 (± 4.416)
% Data Referral	30.578 (± 6.638)	37.200 (± 6.573)	37.000 (± 4.416)
(b)			
Metrics	MCD	DE	EMCD
$\mu Accuracy$	0.907 (± 0.021)	0.923 (± 0.027)	0.933 (± 0.015)
$\mu F1$ score	0.904 (± 0.022)	0.921 (± 0.027)	0.931 (± 0.016)
$\mu U_{Accuracy}$	0.815 (± 0.030)	0.795 (± 0.049)	0.796 (± 0.021)
$\mu U_{Precision}$	0.405 (± 0.104)	0.365 (± 0.054)	0.339 (± 0.092)
μU_{Recall}	0.486 (± 0.100)	0.585 (± 0.093)	0.600 (± 0.098)
$\mu U_{Specificity}$	0.872 (± 0.035)	0.830 (± 0.053)	0.824 (± 0.017)
% Data Retained	81.711 (± 3.946)	77.200 (± 5.020)	76.600 (± 2.302)
% Data Referral	18.289 (± 3.946)	22.800 (± 5.020)	23.400 (± 2.302)

predictions can be referred to medical experts. The table demonstrates that both the EMCD and DE models exhibit competitive and higher U_{Recall} than the MCD model, indicating their effectiveness in identifying high-uncertainty incorrect predictions. Moving on to $U_{Precision}$ metrics, which focus on correctness (whether a prediction is correct or not, particularly in the context of false uncertainties), we observe that MCD also outperforms the other models.

Furthermore, our observation indicates that relying solely on $U_{Accuracy}$ can be misleading, particularly in situations where the costs of FU and FC are significantly different. This metric fails to differentiate between types of errors, necessitating additional metrics for a thorough evaluation. A high U_{Recall} could indicate a model that labels too many predictions as uncertain, which might not be practical for decision-making processes. This metric does not measure the correctness of certain predictions, focusing solely on the management of incorrect predictions. Similarly, $U_{Specificity}$ ignores incorrect predictions, providing no information on how well the model identifies or handles them. Prioritising improvement in one measure can inadvertently impact others negatively. For instance, increasing U_{Recall} by marking more predictions as uncertain could decrease $U_{Precision}$. Additionally, the effectiveness and interpretation of these metrics depend on the chosen threshold for uncertainty, highlighting the importance of careful selection and sensitivity analysis.

Overall, these findings underscore the value of uncertainty models in improving the accuracy and reliability of predictions, especially in domains where precise decision-making is critical, such as medicine. Moreover, we have also visualised examples of input test images and

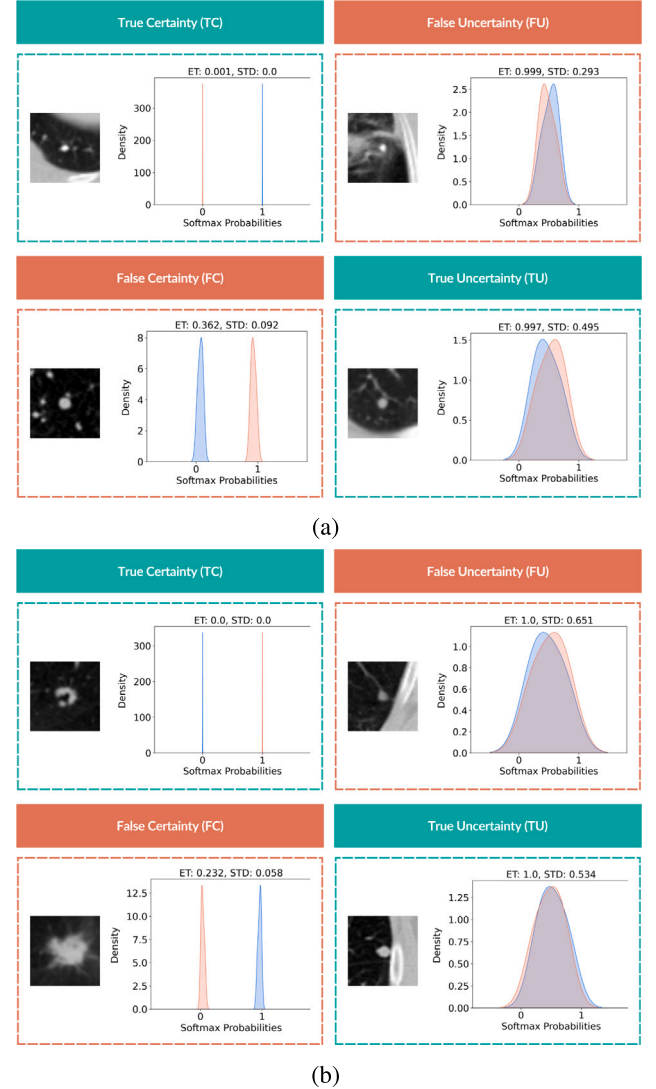


Fig. 9. Distribution of the output posterior distribution for (a) benign nodules and (b) malignant nodules, with the corresponding uncertainty estimates obtained using DE. Predictions are grouped into four outcomes: TC, TU, FU, and FC at the threshold value, $H_{norm} = 0.5$. Green represents the desired outcome and red represents the undesired outcome.

the corresponding predictive distributions generated by the DE model and grouped them by the criteria from the uncertainty confusion metrics depicted in Fig. 9. The blue indicates benign labelling, whereas red indicates malignant. The wider or overlapping the posterior, the less

Table 6
Comparison of the performance of the uncertainty-aware framework using threshold with some existing classification models.

Reference	Year	Accuracy	AUC	F1 score	Dataset composition	Cross-Validation	Methodology	Uncertainty
Dey et al. [18]	2018	0.904	0.955	N/A	686 nodules (46% benign, 54% malignant)	5-fold	3D DenseNet with deep learning techniques	No
Shaffie et al. [58]	2018	0.912	N/A	N/A	727 nodules (413 benign, 314 malignant)	10-fold	Feature analysis with autoencoder approach	No
Xie et al. [21]	2019	0.916	0.957	N/A	1945 nodules (1301 benign, 644 malignant)	10-fold	Knowledge-based collaborative deep learning	No
Sahu et al. [59]	2019	0.932	N/A	N/A	1305 nodules (857 benign, 448 malignant)	10-fold	Lightweight CNN for multiple sections	No
Xiao et al. [22]	2020	0.931	N/A	N/A	4578 nodules	10-fold	CNN with denoising autoencoder and handcrafted features	No
Shen et al. [60]	2021	0.865	0.972	N/A	2305 nodules (1375 benign, 930 malignant)	10-fold	Multi-view CNN from 2D CT views	No
Xia et al. [61]	2021	0.919	N/A	0.910	1001 nodules (554 benign, 447 malignant)	10-fold	Network ensemble with attention mechanisms	No
Al-Shabi et al. [62]	2021	0.953	0.981	N/A	848 nodules (442 benign, 406 malignant)	10-fold	ProCAN with channel-wise attention	No
Tang et al. [63]	2021	0.944	0.931	N/A	1568 nodules (918 benign, 650 malignant)	5-fold	3D Semi-supervised clustering	No
Astaraki et al. [19]	2021	N/A	0.936	N/A	1297 nodule (876 benign, 421 malignant)	5-fold	Dual pathway approach	No
Muzammil et al. [23]	2021	0.969	0.992	N/A	1186 nodules	5-fold	Fusion techniques with SVM and AdaBoost	No
Huang et al. [20]	2022	0.921	0.955	0.899	788 nodules (319 benign, 469 malignant)	5-fold	Optimisation framework for deep features	No
Our study (Th = 0.50)	2023	0.959	N/A	0.958	1013 nodules (561 benign, 452 malignant)	5-fold	3D InceptionResNetV2 with EMCD	Yes
Our study (Th = 0.75)		0.933		0.931				

confident the model or the uncertainty of the prediction. As an example, taking a sample from the TU, Fig. 9(a), the entropy (ET) is 0.997 and the standard deviation (STD) is 0.495 with a wide distribution and the prediction is incorrect. Thus, the model can serve as an informative indicator for detecting such predictions and referring them to medical experts.

4.5. Performance comparison with existing methods

To underscore the performance of our UQ model, we conducted a comprehensive comparison with the existing architectural models used for the classification of lung cancer nodules based on the same dataset, LIDC-IDRI, as outlined in Table 6. Our findings indicate that our model performs at a comparable level to other established models. Upon closer examination, it becomes clear that a significant portion of the pre-existing models employed for lung cancer classification do not account for the uncertainty inherent in the model’s output. The integration of UQ in our model emphasises the advantage of referring uncertain images to medical experts for further evaluation. By employing three different UQ methods, we were able to thoroughly investigate which of these models performs most effectively.

However, it is important to acknowledge that our model, despite its numerous advantages, has potential trade-offs. One of these trade-offs is the extended inference time and computational demands. This is because our model requires multiple evaluations using MC methods, which inherently consume more time. Additionally, the DE model, in particular, necessitates the combination of multiple models. Nevertheless, recent technological advancements, such as high-performance GPUs and large memory storage capacity, have made it increasingly feasible to address these computational challenges.

5. Conclusion

In this study, we present an uncertainty-aware framework for lung nodule classification based on the LIDC-IDRI dataset that employs MCD, DE, and EMCD for UQ. Our findings indicate that model performance improves when uncertain predictions are either referred for further review or discarded. We added a dropout layer just before the classification layer to facilitate MC inference and used three sets of models for ensembling. The results showed that InceptionResNetV2 is

the most promising model among the other seven models, with an F1 score of 0.845, which was further improved to 0.866 using EMCD. EMCD also demonstrated superior accuracy, with scores of 0.959 and 0.933 when using threshold values of 0.5 and 0.75, respectively, underscoring its potential for clinical applications. Further validation of the robustness and reliability of the model was conducted on the SynapseMNIST3D and VesselMNIST3D datasets, where the ensemble approach demonstrated better performance than MCD.

However, while MCD performed well across most uncertainty evaluation metrics, EMCD and DE proved to be more effective at identifying incorrect predictions with higher uncertainty levels. Future work will explore alternative UQ methods, such as DropConnect [48] and Drop-Weights [64], and their integration into the model. We also plan to implement decision-making using multi-criteria decision analysis (MCDA) [65,66] and investigate a feedback mechanism for improving UQ.

CRediT authorship contribution statement

Rahimi Zahari: Conceptualization, Data curation, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Julie Cox:** Supervision, Writing – review & editing. **Boguslaw Obara:** Conceptualization, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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